

Exercise 1: (6 Points) Decide for each of the following statements whether they are true or false. *Hint:* You will get points for correctly answered questions and points subtracted for wrong answers. However, you can not achieve negative points in a given question block.

a) Block I (2 Points)

- i) In linear regression, if the coefficient of a variable is positive, then there must be a positive correlation between that variable and the response variable.
☐ TRUE ☐ FALSE
- ii) In a linear regression model, the least squares estimator is BLUE (Best Linear Unbiased Estimator) if and only if the error term is expected to be zero and has constant variance (i.e., satisfies homoskedasticity).
☐ TRUE ☐ FALSE
- iii) In multiple linear regression, the full model F test is based on the assumption $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$, which follows the F distribution $F \sim F_{k, n-k-1}$.
☐ TRUE ☐ FALSE
- iv) For nominal variables with m categories, we need to add m dummy variables to represent this predictor when constructing the linear model.
☐ TRUE ☐ FALSE

FFTF

b) Block II (2 Points)

- i) When highly correlated explanatory variables are present in a linear regression model, this may lead to a reduction in the standard errors of the model coefficient estimates.
☐ TRUE ☐ FALSE
- ii) When the residuals of a linear regression model are not independent, their correlation does not affect the consistency of the coefficient estimates, but it may affect the efficiency of the estimates.
☐ TRUE ☐ FALSE
- iii) In a linear regression model, when the confidence interval of an explanatory variable contains zero, we cannot reject the hypothesis that the coefficient of that variable is zero.
☐ TRUE ☐ FALSE
- iv) In a well-fitted linear regression model, the mean of the residuals should be close to zero.
☐ TRUE ☐ FALSE

FTTT

c) Block III (2 Points)

- i) The residual sum of squares (RSS) of a linear regression model may increase as more variables are added to the model.
☐ TRUE ☐ FALSE
- ii) When a predictor variable is added to a regression model that is not related to the response variable, the AIC value of the model must decrease.
☐ TRUE ☐ FALSE
- iii) In a linear regression model with n observations and p predictor variables, a degree of freedom $n - p - 1$ t distribution was used to assess the significance of each coefficient.
☐ TRUE ☐ FALSE
- iv) In a linear regression model applying logarithmic transformation, the coefficients on the predictor variables should be interpreted as the percentage change in the response variable for a 1% change in the predictor variable.
☐ TRUE ☐ FALSE

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Exercise 2: (6 Points)

Consider the simple linear regression model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

- a) Briefly describe the main contents of the Gaussian Markov-Theorem and the assumptions. (4 Points)
1 points for every assumption.

- b) Explain how the ordinary least squares (OLS) method is used to estimate the coefficients β_0 and β_1 in the linear regression model. (2 Points)

1 point for correctly state that RSS needs to be minimized. And 1 point for correctly derive the solution.
(Or at least say how the solution needs to be derived)

Exercise 3: (18 Points)

The following Table 1 provides data from a sample of 5 pairs of values (X, Y) . Each pair represents a data point and the sample size is 5.

Table 1: Sample Data of Pairs (X, Y)

X	20	30	33	40	15
Y	7	9	8	11	5

- a) Build a regression model of consumption Y on income X with intercept and estimate the intercept and slope. How do you interpret the estimated intercept and slope? (3 Points)

Hint: Assume $(X'X)^{-1} = \begin{pmatrix} 2.079 & -0.068 \\ -0.068 & 0.0024 \end{pmatrix}$

Solution: The regression should be stated as

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

1. Write down the matrix X , which is

$$X = \begin{pmatrix} 1 & 20 \\ 1 & 30 \\ 1 & 33 \\ 1 & 40 \\ 1 & 15 \end{pmatrix}$$

2. Step by step using OLS calculate the coefficients.

$$X'X = \begin{pmatrix} n & \sum_{i=1}^n x_i \\ \sum_{i=1}^n x_i & \sum_{i=1}^n x_i^2 \end{pmatrix} = \begin{pmatrix} 5 & 138 \\ 138 & 4214 \end{pmatrix}$$

$$(X'X)^{-1} = \frac{1}{\det} \begin{pmatrix} \sum_{i=1}^n x_i^2 & -\sum_{i=1}^n x_i \\ -\sum_{i=1}^n x_i & n \end{pmatrix} = \begin{pmatrix} 2.079 & -0.068 \\ -0.068 & 0.0024 \end{pmatrix}$$

$$X'Y = \begin{pmatrix} \sum_{i=1}^n y_i \\ \sum_{i=1}^n x_i y_i \end{pmatrix} = \begin{pmatrix} 40 \\ 1189 \end{pmatrix}$$

$$\hat{\beta} = (X'X)^{-1}X'Y = \begin{pmatrix} 2.079 & -0.068 \\ -0.068 & 0.0024 \end{pmatrix} \begin{pmatrix} 40 \\ 1189 \end{pmatrix} = \begin{pmatrix} 2.21 \\ 0.209 \end{pmatrix}$$

Therefore, the estimated regression model is

$$\hat{y} = 2.21 + 0.209X$$

One could also use the simple linear regression calculation, which will give the same answer:

$$\begin{aligned} \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \end{aligned}$$

- b) Using the above data, calculate the residual sum of squares (RSS) $\hat{\epsilon}'\hat{\epsilon}$ and the estimate variance $\hat{\sigma}^2$. (2 Points)

Solution:

Given the above estimated model, one can calculate the predicted values of y by

$$\hat{y} = X\hat{\beta} = \begin{pmatrix} 1 & 20 \\ 1 & 30 \\ 1 & 33 \\ 1 & 40 \\ 1 & 15 \end{pmatrix} \begin{pmatrix} 2.21 \\ 0.209 \end{pmatrix} = \begin{pmatrix} 6.40 \\ 8.50 \\ 9.13 \\ 10.60 \\ 5.35 \end{pmatrix}$$

Residuals are

$$\hat{\epsilon} = y - \hat{y} = \begin{pmatrix} 0.594 \\ 0.496 \\ -1.132 \\ 0.398 \\ -0.356 \end{pmatrix}$$

SSR:

$$\hat{\epsilon}'\hat{\epsilon} = 2.169$$

Estimation of variance:

$$\sigma^2 = \frac{\hat{\epsilon}'\hat{\epsilon}}{n-2} = \frac{2.169}{3} = 0.723$$

- c) Conduct a significance test for the slope β_1 . Please clearly write the null and alternative hypotheses, your test statistic, degrees of freedom, critical value, and draw the correct conclusion based on your calculation. You may find the information on page 9 (Tables 2-4) useful. Assume $\alpha = 0.05$. (2 Points)

Solution:

To conduct the hypothesis test, one needs to first estimate the variance and standard error of the parameter $\hat{\beta}_1$, which can be calculated by:

$$\text{var}(\hat{\beta}) = \text{diag}(\hat{\sigma}^2(X'X)^{-1}) = (2.079 * 0.723, 0.0024 * 0.723) = (1.504, 0.084)$$

$$\text{sd}(\hat{\beta}) = (1.226, 0.042)$$

Null hypothesis: $H_0 : \beta_1 = 0$ Alternative hypothesis: $H_1 : \beta_1 \neq 0$.

Test distribution: t-distribution

$$\text{t-statistic: } t = \frac{\hat{\beta}_j}{\text{se}(\hat{\beta}_j)} = \frac{0.20977}{0.04224} = 4.966$$

Degrees of freedom: $n - p - 1 = 5 - 1 - 1 = 3$

Critical value: $t_{0.975,42} = 3.182$ (Here if the student is confused to use other degrees of freedom or other significance level, -0.5)

Conclusion: Since it's a two-sided test, the absolute value of the test statistic is far larger than the critical value. We reject the null hypothesis. The coefficient under a 0.05 significance level is significantly different from 0.

- d) Calculate the 95%-confidence interval for β_1 and interpret the meaning of the confidence interval. (1 Points)

Solution:

Since the standard error is calculated above, we simply calculate the confidence interval using: Formular:

$$CI = \hat{\beta}_j \pm t_{\alpha/2, df} \cdot se(\hat{\beta}_j)$$

$$\text{Results: } 0.20977 \pm 0.04224 \cdot t_{0.975,3} = 0.20977 \pm 0.04224 \cdot 3.182 = [0.07533389, 0.344212]. (1pt)$$

- e) Given that you've already obtained the RSS, calculate the total sum of squares (TSS) and explained sum of squares (ESS), then calculate the coefficient of determination R^2 . How do you interpret the obtained R^2 ? (2 Points)

Solution:

$$TSS = \sum_{i=1}^2 (y_i - \bar{y})^2 = 20$$

$$ESS = TSS - RSS = 20 - 2.169299 = 17.8307$$

$$R^2 = 17.8307/20 = 0.891535 (1pt)$$

R^2 is interpreted as how much of the variation of the dependent variable can be explained by the variation of the model. In this case, we could say that about 89.15% variation of y is explained by the change of X . (1pt)

Exercise 4: The following are extracts from the output of a fitted linear regression model in R:

Call:

```
lm(formula = mpg ~ horsepower, data = Auto)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.5710	-3.2592	-0.3435	2.7630	16.9240

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	39.935861	0.717499	55.66	<2e-16 ***
horsepower	-0.157845	0.006446	-24.49	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.906 on 390 degrees of freedom

Multiple R-squared: 0.6059, Adjusted R-squared: 0.6049

F-statistic: 599.7 on 1 and 390 DF, p-value: < 2.2e-16

- a) The residual diagnostic plots are presented in Figure 1 on page 8. Briefly discuss three problems that the model may have. (3 Points)

Solution:

1. Heteroscedasticity
 2. Non-linearity
 3. Non-normality
- Every explanation for 1 point, if only stated the problem without explanation, then - 0.5 for each problem.

- b) One colleague suggests that instead of fitting the model

$$mpg_i = \beta_0 + \beta_1 horsepower_i + \varepsilon_i, \quad (1)$$

you should fit the regression as

$$mpg_i = \beta_0 + \beta_1 horsepower_i + \beta_2 horsepower_i^2 + \varepsilon_i, \quad (2)$$

which includes the second-order polynomial term. Do you think this proposal can solve the problem you encountered in question (a)? If so, which problem would it solve and why? If not, why? (2 Points)

Solution:

The problem this could solve is the problem of non-linearity in the first question.

- c) To verify the proposal of the colleague in part (a), you calculated the AIC values of the two models and these values are

$$\text{AIC}(\text{Model}_1) = 2363.324$$

$$\text{AIC}(\text{Model}_2) = 2274.354$$

Briefly discuss the meaning and the purpose of AIC here and state which model do you think is better and why. (2 Points)

Solution:

AIC is a measure of how well the model is fitted, after a new variable is added in. If the value of AIC decreases, then it means that the decrease of RSS of the newly added variable on fitting is greater than the penalising effect it brings, and therefore the fitting is better. Generally, we need to pick the model with lower AIC, in this case, the second model has lower AIC and better fitting.

- d) In addition to AIC or BIC, suggest another method that could be used to verify your colleague's conjecture and explain your suggestion. You may consider using some sort of numerical measure or some other scheme, as long as it is reasonable. (1 Point)

Solution:

One could use the F-test, test for significance of the polynomial term, Mallows CP value, predicted MSE, or simply observe the diagnostic plot to see if the fitting is improved.

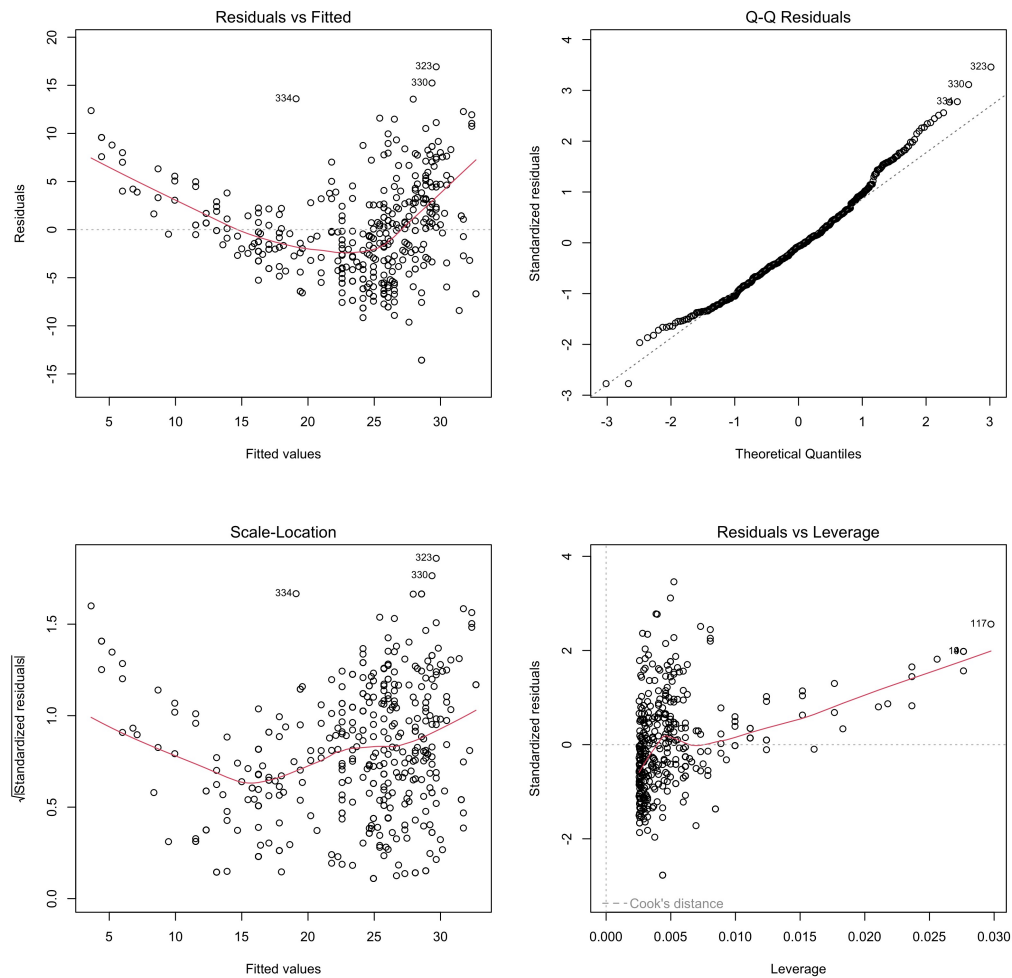


Figure 1: Residual diagnostic plots of the regression.

Quantile tables

Table 2: Quantiles for some t -distributions						
quantile	0.495	0.9	0.95	0.975	0.99	0.995
3	-0.0136	1.6377	2.3534	3.1824	4.5407	5.8409
7	-0.0130	1.4149	1.8946	2.3646	2.9980	3.4995
12	-0.0128	1.3562	1.7823	2.1788	2.6810	3.0545
13	-0.0128	1.3502	1.7709	2.1604	2.6503	3.0123
25	-0.0126	1.3163	1.7081	2.0595	2.4851	2.7874
42	-0.0126	1.3020	1.6819	2.0180	2.4184	2.6980
49	-0.0126	1.2990	1.6766	2.0095	2.4048	2.6799
50	-0.0126	1.2987	1.6759	2.0086	2.4033	2.6778
100	-0.0126	1.2901	1.6602	1.9840	2.3642	2.6259

Table 3: Quantiles for the standard normal distribution						
quantile	0.495	0.9	0.95	0.975	0.99	0.995
	-0.0125	1.2816	1.6449	1.9600	2.3263	2.5758

Table 4: Quantiles for some F -distributions with fixed $df_2 = 49$						
quantile	0.495	0.9	0.95	0.975	0.99	0.995
$df_1 = 3$	0.7905	2.1991	2.7939	3.3961	4.2084	4.8379
4	0.8425	2.0633	2.5611	3.0602	3.7283	4.2430
5	0.8747	1.9685	2.4044	2.8384	3.4162	3.8596
6	0.8965	1.8980	2.2904	2.6793	3.1948	3.5893
7	0.9123	1.8431	2.2032	2.5587	3.0285	3.3871
8	0.9246	1.7989	2.1340	2.4637	2.8983	3.2294
9	0.9336	1.7625	2.0775	2.3866	2.7932	3.1025
10	0.9411	1.7319	2.0303	2.3226	2.7064	2.9979

Table 5: Quantiles for some F -distributions with fixed $df_2 = 42$						
quantile	0.495	0.9	0.95	0.975	0.99	0.995
$df_1 = 3$	0.7924	2.2191	2.8270	3.4457	4.2853	4.9396
4	0.8444	2.0837	2.5943	3.1089	3.8021	4.3394
5	0.8766	1.9894	2.4377	2.8866	3.4882	3.9528
6	0.8985	1.9193	2.3240	2.7273	3.2658	3.6804
7	0.9143	1.8649	2.2371	2.6068	3.0988	3.4768
8	0.9263	1.8211	2.1681	2.5118	2.9681	3.3181
9	0.9356	1.7850	2.1119	2.4348	2.8628	3.1906
10	0.9432	1.7547	2.0650	2.3710	2.7758	3.0855